Run report: item1D-run-20261018T2056Z

Descriptive record of one recorded run of examples/decomposition_coverage.rs (Part D) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
run: item1D, 20261018T2056Z
protocol commits: 55ebe41b6 (Addendum A), Amendment B (Poisson residual, frozen before this run); tool commit: fe0469c07
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz, 15 GB
command: rustc --edition 2021 -O examples/decomposition_coverage.rs -o tool && DC_THREADS=2 ./tool research/weil_descent_angles_20261018/results/item1D-run-20261018T2056Z
tool sha256: e4e38c882b23b23fc72750b47779e8113e7827f3a092a8d9b6d7ad3d406e0a73
threads: 2 (sharing the host with the item-4 census and the other coverage run; outputs are exact counts, no timing is evidence)
smoke runs before this run (not evidence): DC_SCALE=0.1 / PKM_SCALE=0.15, single thread.
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

tool_log.txt

```
jobs: 273
D char 0 n=2: curves 672 lambda 0.5033 | pi subfield 0.3949 null 0.3931 positive 0.0053 1/n! 0.5000 | resid subfield +0.00017 null +0.00001 diff +0.00016 (z 0.94)
D char 0 n=3: curves 63 lambda 0.1864 | pi subfield 0.1576 null 0.1787 positive 0.0029 1/n! 0.1667 | resid subfield -0.00985 null -0.00977 diff -0.00007 (z -0.10)
D char 0 n=4: curves 21 lambda 0.0831 | pi subfield 0.0605 null 0.0625 positive 0.0073 1/n! 0.0417 | resid subfield -0.01779 null -0.01691 diff -0.00088 (z -0.19)
D char 0 n=5: curves 9 lambda 0.0369 | pi subfield 0.0210 null 0.0187 positive 0.0032 1/n! 0.0083 | resid subfield -0.01439 null -0.01174 diff -0.00265 (z -0.43)
D char 2 n=2: curves 21 lambda 0.5073 | pi subfield 0.3730 null 0.4054 positive 0.0296 1/n! 0.5000 | resid subfield -0.02465 null +0.00615 diff -0.03080 (z -4.72)
D char 2 n=3: curves 15 lambda 0.1683 | pi subfield 0.1385 null 0.2039 positive 0.0065 1/n! 0.1667 | resid subfield -0.01150 null -0.02743 diff +0.01592 (z 1.96)
D char 2 n=4: curves 9 lambda 0.1467 | pi subfield 0.0818 null 0.0711 positive 0.0037 1/n! 0.0417 | resid subfield -0.03826 null -0.02154 diff -0.01672 (z -1.06)
D char 2 n=5: curves 9 lambda 0.0493 | pi subfield 0.0265 null 0.0529 positive 0.0057 1/n! 0.0083 | resid subfield -0.02066 null -0.04763 diff +0.02696 (z 1.11)
total 134.0s
```

checks.txt

```
D char 0 n=2: curves 672 lambda 0.5033 | pi subfield 0.3949 null 0.3931 positive 0.0053 1/n! 0.5000 | resid subfield +0.00017 null +0.00001 diff +0.00016 (z 0.94)
D char 0 n=3: curves 63 lambda 0.1864 | pi subfield 0.1576 null 0.1787 positive 0.0029 1/n! 0.1667 | resid subfield -0.00985 null -0.00977 diff -0.00007 (z -0.10)
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D char 2 n=5: curves 9 lambda 0.0493 | pi subfield 0.0265 null 0.0529 positive 0.0057 1/n! 0.0083 | resid subfield -0.02066 null -0.04763 diff +0.02696 (z 1.11)
```

coverage.csv (first 12 of 2457 rows)

char	n	q	curve	order	arm	x_values	F	lambda	poisson	inv_n_factorial	distinct	multisets	pi
0	2	53	0	2749	subfield	53	56	0.580575	0.440423	0.500000	1221	1596	0.444162
0	2	53	0	2749	null	53	64	0.756639	0.530759	0.500000	1411	2080	0.513278
0	2	53	0	2749	positive	53	56	0.580575	0.440423	0.500000	113	1596	0.041106
0	2	53	1	2746	subfield	53	50	0.464312	0.371432	0.500000	1007	1275	0.366715
0	2	53	1	2746	null	53	54	0.540787	0.417710	0.500000	1129	1485	0.411143
0	2	53	1	2746	positive	53	50	0.464312	0.371432	0.500000	101	1275	0.036781
0	2	53	2	2864	subfield	53	52	0.481145	0.381925	0.500000	1121	1378	0.391411
0	2	53	2	2864	null	53	36	0.232542	0.207483	0.500000	580	666	0.202514
0	2	53	2	2864	positive	53	52	0.481145	0.381925	0.500000	105	1378	0.036662
0	2	59	0	3552	subfield	59	54	0.418074	0.341687	0.500000	1208	1485	0.340090
0	2	59	0	3552	null	59	72	0.739865	0.522822	0.500000	1888	2628	0.531532
0	2	59	0	3552	positive	59	54	0.418074	0.341687	0.500000	109	1485	0.030687

coverage_summary.csv (first 8 of 8 rows)

char	n	curves	mean_lambda_subfield	mean_pi_subfield	mean_pi_null	mean_pi_positive	inv_n_factorial	resid_subfield	resid_null	resid_diff	sd_resid_diff	z_resid_diff
0	2	672	0.503345	0.394904	0.393134	0.005322	0.500000	0.000170	0.000012	0.000158	0.004338	0.94
0	3	63	0.186410	0.157583	0.178683	0.002853	0.166667	−0.009846	−0.009773	−0.000073	0.005621	−0.10
0	4	21	0.083102	0.060455	0.062501	0.007257	0.041667	−0.017793	−0.016909	−0.000883	0.020966	−0.19
0	5	9	0.036891	0.020982	0.018669	0.003245	0.008333	−0.014387	−0.011739	−0.002649	0.018667	−0.43
2	2	21	0.507316	0.372989	0.405361	0.029629	0.500000	−0.024652	0.006151	−0.030803	0.029887	−4.72
2	3	15	0.168324	0.138491	0.203864	0.006471	0.166667	−0.011502	−0.027426	0.015924	0.031511	1.96
2	4	9	0.146729	0.081766	0.071056	0.003683	0.041667	−0.038259	−0.021542	−0.016717	0.047470	−1.06
2	5	9	0.049294	0.026534	0.052920	0.005659	0.008333	−0.020663	−0.047627	0.026964	0.073180	1.11

Graph

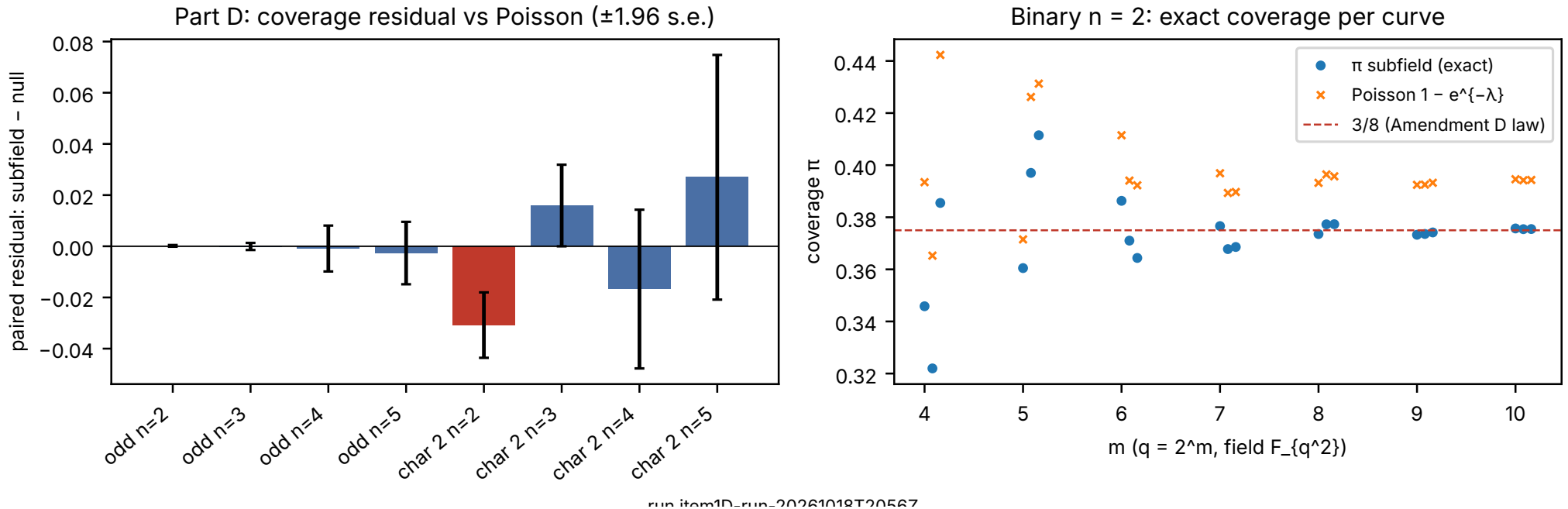


Figure 1: Part D: paired residual from Poisson per (characteristic, n), and binary n = 2 coverage per curve; exact.